

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, **sentence-transformers**, React), productionizing a published NLP method, embedding validated items and corpus texts with the same model and scoring by cosine similarity, into a self-serve tool for researchers.
- Built a **YAML model registry** (embedding model, E5 prefix policy, language coverage) as single source of truth, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer integrating AI models into a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend and APIs that build alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to structured synthesis while the backend deterministically controls eligibility and coverage.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack ML training-data annotation platform (React, Node.js, **MongoDB**) used daily by the AI team, building data pipelines that cut manual annotation by **100+ hours/month** and lifted labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, **MongoDB**) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; routed OpenAI/Groq with fallback and hardened it with **150+ tests**.

align-ops – LLM Post-Training + Serving Ops PyTorch, TRL (SFT/DPO/GRPO), vLLM, FastAPI, Docker

- Built an end-to-end LLM post-training and serving pipeline (SFT, DPO, GRPO via TRL with QLoRA) that lifted held-out task accuracy **54% to 72%** and won **71% of 500** preference evaluations over the base model, served behind a FastAPI + vLLM gateway with Redis caching and async micro-batching (**1.9x** throughput) and Prometheus/Grafana monitoring.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR over sentence-transformer embeddings), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & DB: Python, TypeScript, JavaScript, SQL, MongoDB, PostgreSQL, Redis, SQLite

ML & AI: PyTorch, sentence-transformers, Hugging Face Transformers, scikit-learn, JAX, NumPy, pandas, model evaluation

LLM & RAG: LLM agents, RAG, embeddings, OpenAI / Groq / Gemini APIs, RRF, cross-encoder reranking, MMR, provenance

Backend & APIs: FastAPI, Node.js, Express, REST APIs, async services, Git, GitHub Actions, pytest

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Firebase, Cloud Run, Linux, MongoDB Atlas

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.